

Lesson 14 — Nested Loops (+ Debug)

Matches LEGO Teacher Portal lesson: "Slide Signal".

Objective (I can...): I can put one loop inside another to make bigger repeating patterns. **CS concept:** Nested loops · **Builds on:** Lesson 13 (loops and stopping).

Stretch/extension, not required mastery. Nesting is a stretch goal — especially for the youngest students. A single working loop (L11–13) is the required takeaway; treat nesting as enrichment and don't hold a struggling pair to it.

AI Connection (2 min): AI training loops nest — for each round, loop through every example. *First-principles question:* "If the outer loop is 'rounds' and the inner is 'examples,' how many looks total is 3 rounds of 4 examples?"

Materials: LEGO CS & AI kit + device per pair, grid or dot paper per pair. **Model reuse:** Loops model from Lesson 11 — code only.

60-minute flow - Warm-up (8): "Rows and beats" — pairs trace a pattern that repeats within a repeat (3 rows of 4 dots) and count the total. - **Concept + demo (10):** Build an inner loop (beep 3×) inside an outer loop (repeat 2×); run so students hear 2 sets of 3. - **Build/Code (30):** Pairs code a nested loop (spin-beep 3×, all repeated 2×). **Debug mini-task (3 min):** give a program where inner and outer counts are swapped — pairs find why the total is wrong and fix it. *Swap at 15 min.* - **Challenge (7):** Change the inner or outer count; predict the new total before running. - **Share + pack-up (5):** Pairs demo and state total inner runs. Model stays built.

Pair roles: Driver builds the inner loop, Navigator adds the outer and tracks counts. Swap at 15 min. (*Trio: third leads the debug.*)

Differentiation: *Support* — a twin nested example to adapt. *Extension / fast-finisher* — add a third layer or a light in the inner loop. *Reteach* — build and test the inner loop alone first, then wrap the outer around it.

Exit check: Pair runs a working nested loop and can explain inner vs. outer counts. **At-home (optional):** Draw a 2×3 sticker pattern; write it as a loop inside a loop. ~10 min.
